॥ श्रीगणेशाय नमः ॥

Encoding and Decoding

26	25	24	23	22	21	20	19	18	17	16	15	14
A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13

26	25	24	23	22	21	20	19	18	17	16	15	14
Z	Y	X	W	V	U	T	S	R	Q	P	O	N
1	2	3	4	5	6	7	8	9	10	11	12	13

E J O T Y
5 10 15 20 25

Q. MANGO = 19-7-20-13-21. Then DANGER = ?
→ 10-7-20-13-11-24 (+6)

Q. E=10. J=20. O=30 T=40. P+E+S+T = ?
→
E J O T
5 10 15 20
↓ ↓ ↓ ↓ ×2
10 20 30 40

P E S T
16 5 19 20
↓ ↓ ↓ ↓
32 10 38 40

Q. If AT=20 BAT=40 CAT = ?
→
AT = 20
1 20

CAT = 60
3 1 20

BAT = 40
2 1 20

Q. R=6, PEL=11, GAP = ?
→
R=6
18 → ÷3

PEL = 11
16 5 12
sum=33 → ÷3

GAP
7 1 16
sum=24 → ÷3 → 8

Q. AT × EAT = 5. EAR × FEAR = 6 SAT × SEAT = 5. Then ACT × FACT = ? EAR × DEAR ?
→
AT
1 20

EAT
5 1 20

EAR × FEAR
6

SAT × SEAT
5

ACT × FACT = 6

EAR ~ DEAR = 4

Q. PLANT = 12·60, BARS = 10, DEEZ = ? GATE = ?
→
PLANT
16 12 1 14 20
sum=63
↓ ÷5
12·6

BARS
2 1 18 19
40
↓ ÷4
10

DEEZ
4 5 5 26
40
↓ ÷4
10

GATE
7 1 20 5
33 ÷ 4
8·25

Q. If ACT = 23-25-16, BLOW = ?
→
ACT
1 3 20
+22 +22 +22
23 25 42 %26
16

BLOW
2 12 15 23
+22 +22 +22 +22
24 34 37 45
↓ ↓ ↓
8 11 19

Q. CONTRACT = 24 STUDENT = 21 GROWTH = ?

→ CONTRACT: 8 ×3→ 24

STUDENT: 7 ×3→ 21

GROWTH: 6 ×3 = 18

Q. CONTRACT = 56, STUDENT = 42, VENKY

→ 8 ×7

7 ×6

VENKY: 5 ×4 20

Q. 24+18 = 54 52+18 = 63 43+25 = 49 15+61 = ?

→ 24+18 → 6 × 9 = 54

52+18 → 7 × 9 = 63

43+25 → 7 7 = 49

15+61 → 6 7 = 42

Q.

24 + 18 = 64	15+61 = ?
52 + 18 = 80	
43 + 25 = 120	

30

Q. 11×12×13 = 234 → 367 (sum of digits)

23×45×16 = 597

12×15×16 = ?

Q. 5+3+2 = 153010 → 5+3+2 = 15 30 10 (a×b a×b×c a+b+c)

2+4+10 = 88016

7+5+6 = ? 7+5+6 = 35 210 18

Q. TOGETHER = RSEGRTCT PAROLE = ?

→ T O G E T H E R

-2 +2 -2 +2 -2

R S E G R

P A R O L E

-2 +2 -2 +2 -2 +2

N C P S J G

Q. CHAMPION = HCMAIPNO NEGATIVE = ?

→ C H A M P I O N

H C M A I P N O

N E G A T I V E

E N A G I T E V

Q. DELHI = CCIDD, BOMBAY

→ 4 5 12 8 9

D E L H I

-1 -2 -3 -4 -5

C C I D D

3 3 9 4 4

2 15 13 2 1 25

B O M B A Y

-1 -2 -3 -4 -5 -6

A M J X V S

Q. SWITCH = TVJSDG, ___ = CGFZE ?

→ S W I T CH

+1 -1

T V J S

B R E A D

+1 -1 +1 -1 +1

C G F Z E

Q. ROME → SUNI BOMBAY → CUNCEZ VENKY = ?

→

R	O	M	E
18	15	13	5

S	U	N	I
19	21	14	9

B	O	M	B	A	Y
C	U	N	C	E	Z

V	E	N	K	Y
W	I	O	L	Z

A	S	I	A
E	T	O	E

vowel → next vowel (A→E, E→I, I→O, O→U, U→A)
consonant → next consonant (B→C, D→F)

Q. MONKEY → XDJMNL BOMBAY → XZALNA TIGER = ?

→

(-1)

M	O	N	K	E	Y
X	D	J	M	N	L

T	I	G	E	R
S	D	F	H	S

Q. PROCESSOR → D4 F3 C5 C1 E1 S1 S1 E3 C6, QUADRANT = ?

→

16	18	15	3	5	19	19	15	18
P	R	O	C	E	S	S	O	R

D4	F3	C5	C1	E1	S1	S1	E3	C6
44	63	35	31	51	191	191	53	36

U=21 D=4 R=18

a) Q1 C7 A1 B2 F6 A1 B7 E4
b) Q1 D6 ~~A1 B2 F3 A1 B7 E4~~
c) Q1 C7 A1 B2 F3 A1 B7 E4
d) Q1 C7 A1 D2 C6 ~~A1 B7 E~~

Q. 851 = good sweet fruit Sweet = ?
783 = good red rose
341 = rose and fruit

→

good = 8
fruit = 1
sweet = 5

Q. Kew xas huma deko = she is eating apples.
Kew tepo qua = she sells toys.
su lim deko = I like apples.

She = ?
apples = ?

→

apples = deko
she = Kew

Q. Matrix I:

	0	1	2	3	4
0	P	A	G	R	Z
1	G	R	Z	P	A
2	Z	P	A	G	R
3	A	G	R	Z	P
4	R	Z	P	A	G

Matrix II:

	5	6	7	8	9
5	E	M	L	N	O
6	L	E	O	M	N
7	O	N	E	L	M
8	N	O	M	E	L
9	M	L	N	O	E

A letter from these matrices can be represented by row and column.
e.g. A = 01, 14, etc.
Find set for ORGAN.

a) 86°, 401, 23G, 14A, 96L ✗
b) 98°, 03, 44G, 22A, 58N
c) 75°, 03, 11R ✗, 22A, 76N
d) 67°, 22, 31, 58N ✗, 22 ✗

condition-based encoding :

Q.

LETTER	P	M	K	E	R
digit/ symbol	5	6	#	2	$

A	T	W	I	J
8	%	1	©	7

U	B	F	H	N
@	9	3	4	*

conditions :

① If both the 1st and last letters in the group are vowels, both are NOT to be coded as the code for the last letter.

② If both the 1st and the last letters in the group are consonants, both are to be coded as the code for the 1st letter.

③ If 1st is consonant, last is vowel, codes for 1st & last are interchanged.

Q. A H N R M U F

↓ ↓ ↓ ↓ ↓ ↓ ↓

8 + * $ 6 @ #

No condn satisfied.
code as it is.

Q. B N A W J P I

3rd condition

© * 8 1 7 5 9

Q. Rahul's roll # is a # consisting of 3 non-zero distinct digits s.t sum of the digits at 100th place and unit place is equal to that of 1/2 of the digit at 10's place. Also, the sum of all possible 3 digit #s obtained using these 3 digits w/o repetition is 2664.

a) digit at unit's place?

b) ten's

c) 100's

→ x y z $\quad x+z = \frac{1}{2}y$

x y z
x z y
y x z
y z x
z x y
z y x

→ sum has # of x's = # of y's = # of z's

$= 100 + 100 + 10 + 1 + 1 + 10$

$= 222$

$222(x+y+z) = 2664$

$x+y+z = 12$

$\frac{1}{2}y + y = \frac{3}{2}y = 12$

$y = 8$

$x+z = 4$

Non-zero distinct

So 1 8 3

3 8 1

Q. Δ : is greater than

% : is lesser than

□ : is equal to

= : is not equal to

+ : is a little more than

X : is a little less than

If $a \Delta b$, $b + c$ then

a) a % c

b) c % a

c) c + a

d) can't say

→ $a > b$

$b > c$

$\Rightarrow a > c$

opt. b.

Q. $a \times b$, $b \square c$, then a) $c+a$ b) $b \Delta c$ c) $a+c$ d) $c \square c$

→ $a<b$ $b=c$ $\Rightarrow$ $a<c$ opt. a.

Q. $> \rightarrow +$
$< \rightarrow -$
$+ \rightarrow \div$
$\wedge \rightarrow \times$
$- \rightarrow =$
$\times \rightarrow >$
$= \rightarrow <$

a) $6+3>1=4+2<1$ $\quad 6 \div 3+1 < 4 \div 2-1 \Rightarrow 3 < 1$ ✗

b) $4>6+2\times32+4<1$ $\quad 4+6 \div 2 > 32 \div 4-1 \Rightarrow 7 > 7$ ✗

$8<4+2=6>3$ $\quad 8-4 \div 2 < 6+3 \Rightarrow 6 < 9$ ✓

$14+7>3=6+3>2$ $\quad 14 \div 7+3 < 6 \div 3+2 \Rightarrow 5 < 4$ ✗

Q. Shamita is fond of collecting greeting cards. Her collecⁿ of 211 cards has a good mix of b'day cards, new year cards, Diwali cards, Christmas cards, and even a marriage anniversary card.
The # of b'day cards = sum of all other except the marriage anniversary card.
The # of new year cards is twice the # of Diwali cards, which in turn is twice the # of Christmas cards.

→ b, n, d, c, m (1)

$b = n+d+c$
$n = 2d$
$d = 2c$

$n = 4c$
$b = 4c+2c+c = 7c$

$7c+4c+2c+c+m = 211$ (m = 1)

$14c = 210$
$c = 15$

$b = 105$, $n = 60$, $d = 30$, $c = 15$, $m = 1$